Technical Service Bulletin

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QSK38/QSK50 Oversize Cylinder Liners	

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Warranty Statement

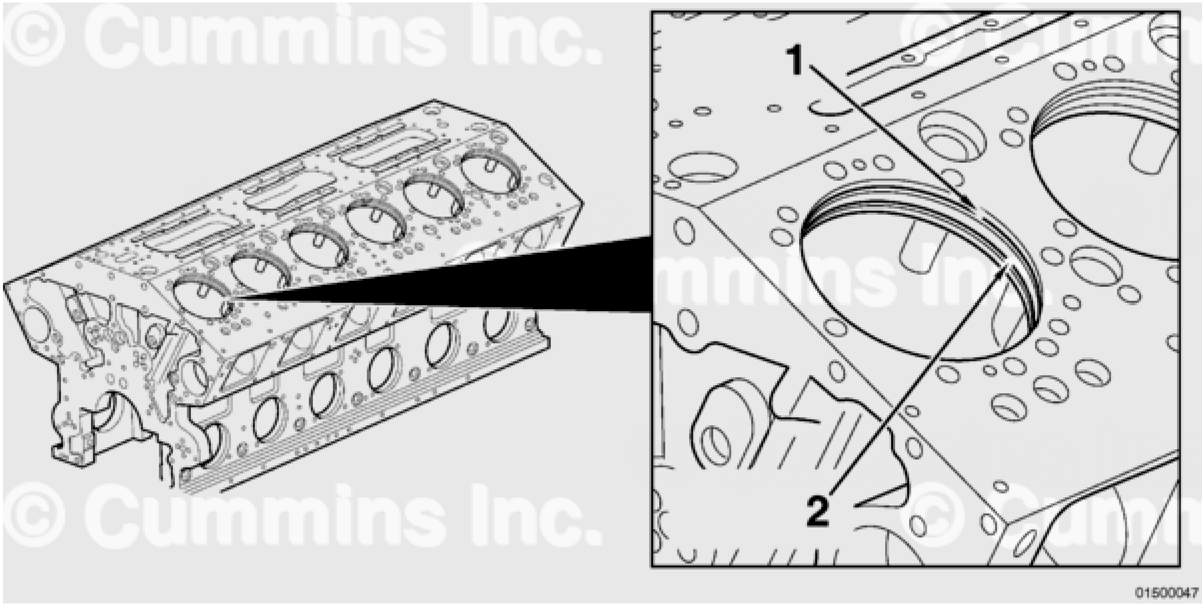
The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

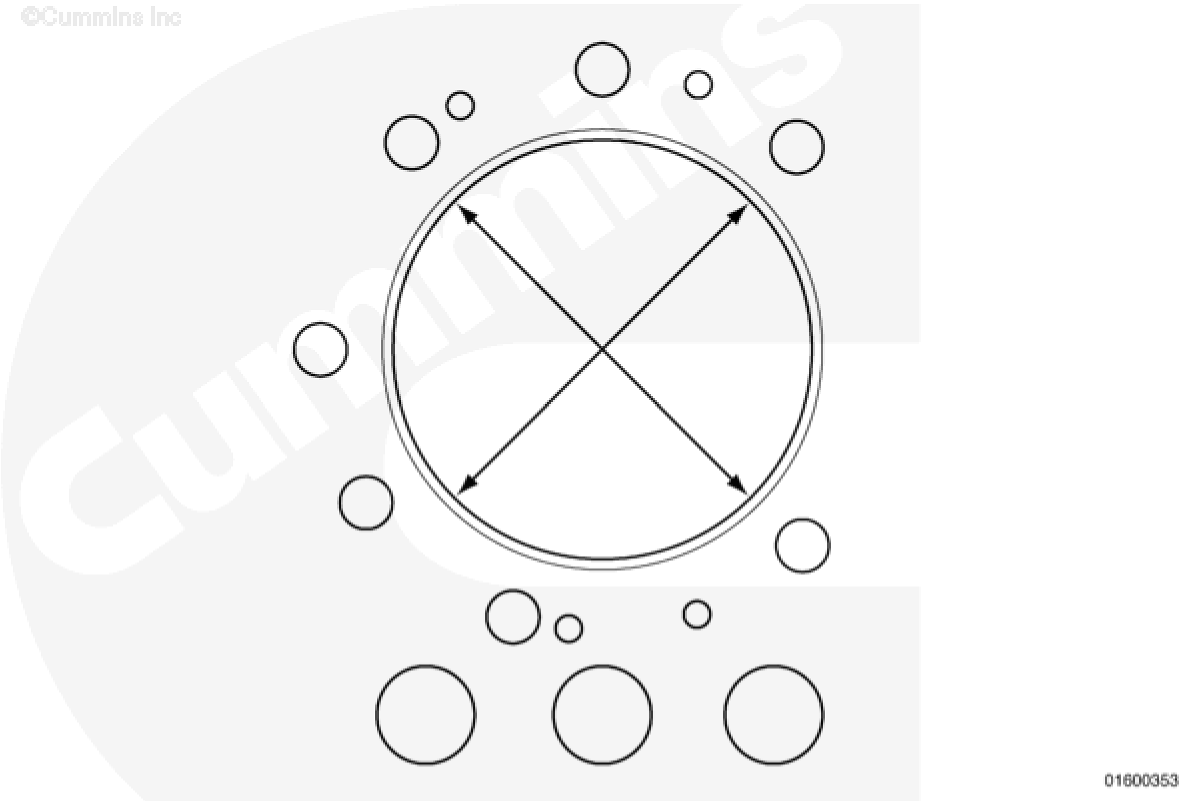
This document informs the field of oversize cylinder liners for use on the QSK38 and QSK50 engines.

The new oversize cylinder liners were released to improve cylinder block serviceability. The oversize dimensional feature of the liner applies to the upper press fit diameter (2) and the upper press fit seating ledge (1) for each oversize.

The oversize cylinder liners are intended for use when there is a requirement to machine the upper/lower press fit and or upper press fit seating ledge due to any wear or fretting damage that may have occurred. During engine rebuild, the upper/lower press fit areas and upper press fit seating ledge of the cylinder block should be inspected for any evidence of wear or fretting. Any cylinder bores that show signs of wear and/or fretting should be machined and an appropriate oversize cylinder liner installed. Engine block upper and lower press fits that show no signs of wear or fretting need **not** have oversize cylinder liners installed and standard cylinder liners can be used.



Shown is the upper press fit diameter (2) location and the upper press fit seating ledge (1) location.



Cylinder block upper and lower press fit bores should be checked for wear across the diameter at the locations shown above. Reference Table 1 for wear specifications.

Table 1: Cylinder Upper and Lower Bore Specifications						
Upper Press Fit				Lower Press Fit		
mm		in		mm		in
190.28	MIN	7.492		181.74	MIN	7.155

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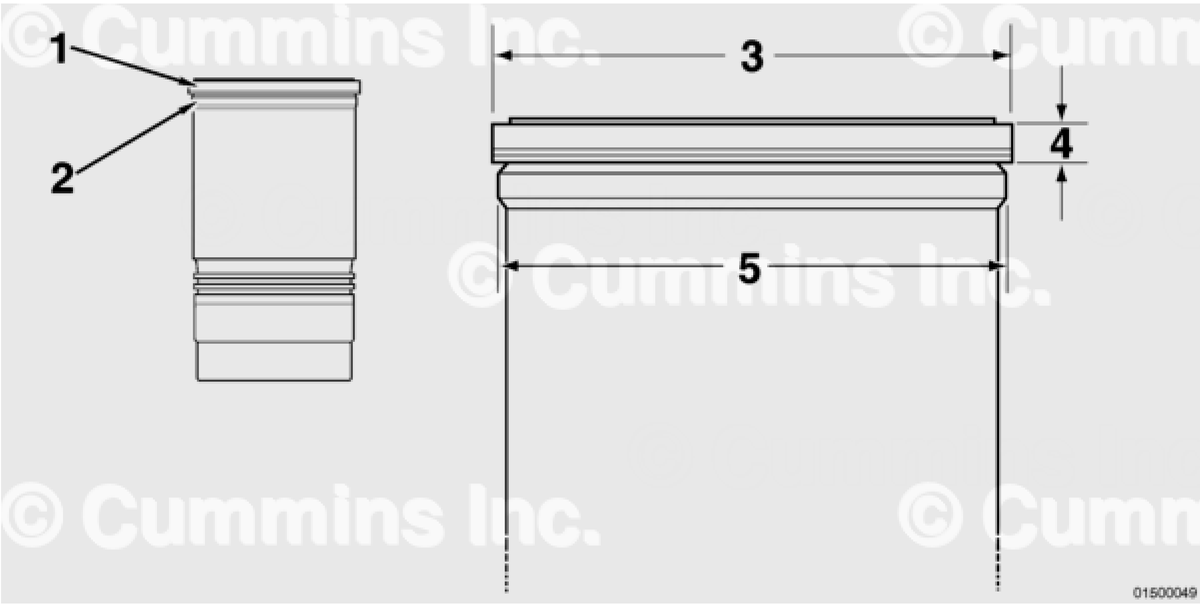
Upper Press Fit				Lower Press Fit		
mm		in		mm		in
190.34	MAX	7.494		181.80	MAX	7.157

Four oversize cylinder liner parts have been released.

- 0.25 mm [0.010 in]
- 0.51 mm [0.020 in]
- 0.76 mm [0.030 in]
- 1.02 mm [0.040 in].

Available cylinder liner shims can be used with both the standard cylinder liner and all oversize cylinder liners. Reference Table 3.



The oversize dimensional features of the liner apply to the upper press fit diameter (1 and 3) and the upper press fit seating ledge (4). All other dimensional features of the cylinder liner remain unchanged.

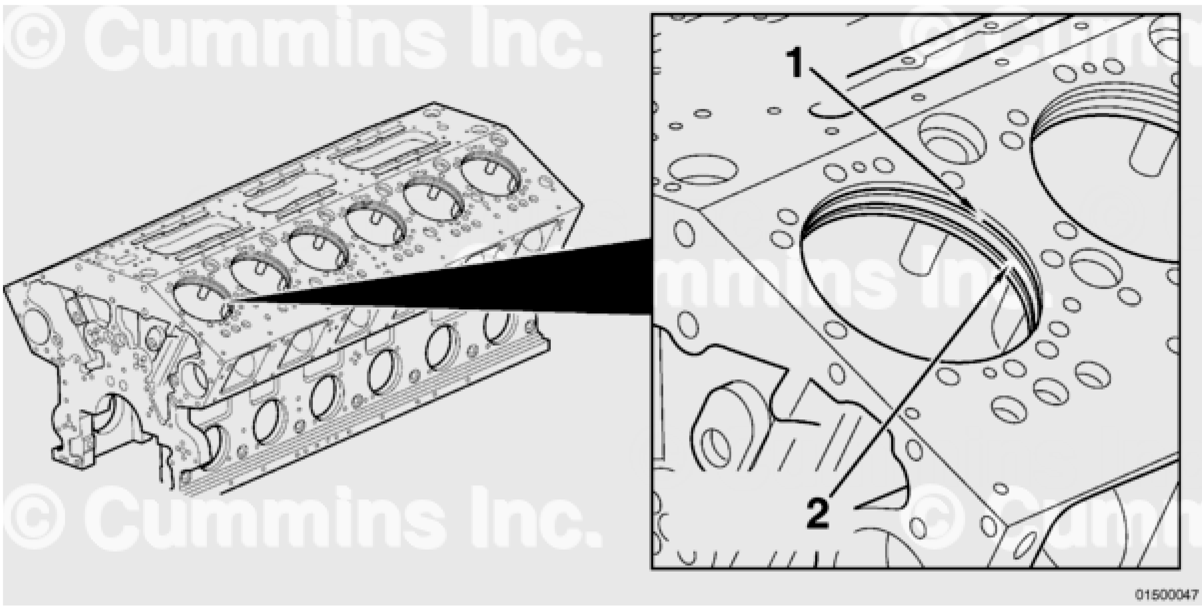
The upper press fit (1 and 3) and the upper press fit seating ledge dimensions (4) all increase by the respective liner oversize value. The 0.25 mm [0.010 in] oversize liner, has the upper press fit and the upper press fit seating ledge dimensions all increased by 0.25 mm [0.010 in] from their nominal dimensional values. Reference Table 2.

Table 2: Cylinder Liner Part Numbers and Key Dimensions

Cylinder Liner Part Numbers and Key Dimensions								
Liner Part Number	**Liner Kit Part Number**	**Oversize Value**	**Upper Press Fit Diameter (3)**		**Lower Press Fit Diameter (5)**		**Upper Press Fit Seating Ledge Depth (4)**	
			MIN	**MAX**	**MIN**	**MAX**	**MIN**	**MAX**
4916452	4955328	Standard Liner	190.31 mm [7.493 in]	190.36 mm [7.495 in]	181.81 mm [7.158 in]	181.86 mm [7.160 in]	13.373 mm [0.526 in]	13.374 mm [0.527 in]
3648481	4352244	0.25 mm [0.010 in]	190.56 mm [7.503 in]	190.61 mm [7.505 in]	181.81 mm [7.158 in]	181.86 mm [7.160 in]	13.627 mm [0.536 in]	13.628 mm [0.537 in]
3648482	4352245	0.51 mm [0.020 in]	190.82 mm [7.513 in]	190.87 mm [7.515 in]	181.81 mm [7.158 in]	181.86 mm [7.160 in]	13.881 mm [0.546 in]	13.882 mm [0.547 in]
3648483	4352246	0.76 mm [0.030 in]	191.07 mm [7.523 in]	191.12 mm [7.525 in]	181.81 mm [7.158 in]	181.86 mm [7.160 in]	14.135 mm [0.556 in]	14.136 mm [0.557 in]
3648484	4352247	1.02 mm [0.040 in]	191.33 mm [7.533 in]	191.38 mm [7.535 in]	181.81 mm [7.158 in]	181.86 mm [7.160 in]	14.389 mm [0.566 in]	14.390 mm [0.567 in]

Cylinder Block Key Machining Dimensions								
For Liner Part Number	**For Liner Kit Part Number**	**Oversize Value**	**Upper Press Fit Diameter (1)**		**Lower Press Fit Diameter (2)**		**Upper Press Fit Seating Ledge Depth in Block**	
			MIN	**MAX**	**MIN**	**MAX**	**MIN**	**MAX**
4916452	4955328	Standard Liner	190.28 mm [7.492 in]	190.34 mm [7.494 in]	181.74 mm [7.155 in]	181.80 mm [7.157 in]	18.237 mm [0.718 in]	18.288 mm [0.720 in]
3648481	4352244	0.25 mm [0.010 in]	190.53 mm [7.502 in]	190.59 mm [7.504 in]	181.74 mm 7.155 in]	181.80 mm [7.157 in]	18.491 mm [0.728 in]	18.542 mm [0.730 in]

Cylinder Block Key Machining Dimensions								
For Liner Part Number	For Liner Kit Part Number	Oversize Value	Upper Press Fit Diameter (1)		Lower Press Fit Diameter (2)		Upper Press Fit Seating Ledge Depth in Block	
			MIN	MAX	MIN	MAX	MIN	MAX
3648482	4352245	0.51 mm [0.020 in]	190.79 mm [7.512 in]	190.85 mm [7.514 in]	181.74 mm [7.155 in]	181.80 mm [7.157 in]	18.745 mm [0.738 in]	18.796 mm [0.740 in]
3648483	4352246	0.76 mm [0.030 in]	191.04 mm [7.522 in]	191.10 mm [7.524 in]	181.74 mm [7.155 in]	181.80 mm [7.157 in]	18.999 mm [0.748 in]	19.050 mm [0.750 in]
3648484	4352247	1.02 mm [0.040 in]	191.30 mm [7.532 in]	191.36 mm [7.534 in]	181.74 mm [7.155 in]	181.80 mm [7.157 in]	19.253 mm [0.758 in]	19.304 mm [0.760 in]



If a standard (non-oversize) liner is to be installed, no work needs to be carried out on the engine block. If an oversize liner is to be installed, the engine block **must** be machined for the respective oversize cylinder liner. The upper press fit (1), lower press fit (2), and upper press fit seating face features **must** all be machined to enable the correct fit of an oversize liner. Any number of cylinder bores can be machined for oversize liners.

Two shim thicknesses are available to enable the correct liner protrusion to be obtained upon engine assembly. Reference Table 3.

Table 3: Shim Part Numbers			
Liner Part Number	**Shim Part Number**	**Shim Thickness**	**Shim Inner Diameter**
4916452 3648481 3648482 3648483 3648484	205744	0.51 mm [0.0200 in]	181.99 +0.127/-0.000 mm [7.165 +0.005/-0.000 in]
	205745	0.79 mm [0.0310 in]	

Four insert ring thicknesses are available to enable the correct liner protrusion to be obtained upon engine assembly. Reference Table 4.

Table 4: Insert Ring Part Numbers			
Liner Part Number	**Insert Ring Part Number**	**Insert Ring Thickness**	**Insert Ring Inner Diameter**
4916452 3648481 3648482 3648483 3648484	3011884	5.067 mm [0.1995 in]	181.91 +0.127/-0.000 mm [7.162 +0.005/-0.000 in]
	3643060	5.093 mm [0.2005 in]	
	3019150	5.118 mm [0.2015 in]	
	3021148	5.169 mm [0.2035 in]	

Document History

Date	Details
2014-6-30	Module Created
2015-12-14	Update to cylinder block upper press fit seating ledge depth in block min and max values.

Last Modified: 15-Dec-2015